

PENETRATION TESTING REPORT

PASSWORD CRACKING USING DIFFERENT TOOLS

W3-PM-FINAL | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B082-Networkwalks
Date	27 September 2026
Modules completed	W3-John The Ripper W3-Using Network Walks Tools
Client/Target	Targeted PDF files owned by Network Walks
Permission secured from client?	Yes
Phases covered	Phase 1: Password cracking using John the Ripper Phase 2: Hash Calculation and Password Cracking using Network Walks Tools

1. Liability Disclaimer

The activities documented herein were performed strictly for authorized educational, testing, and security-hardening purposes. The information, methodologies, and tools detailed in this document are intended solely to demonstrate security vulnerabilities and improve organizational defense mechanisms. The author accepts no liability for any misuse, damage, or illegal activity resulting from the unauthorized application of these techniques or tools outside the scope of authorized testing environments.

2. Introduction

Weak or poorly implemented passwords remain one of the most common vectors for unauthorized data access. Security mechanisms for document encryption, such as those protecting Portable Document Format (PDF) files, are only as strong as the underlying user passwords and the encryption algorithms used.

This assessment evaluates the resilience of password-protected PDF documents against various recovery and cracking methods. By simulating real-world brute-force, dictionary, and automated attacks, this testing aims to identify security gaps in document password policies and evaluate the effectiveness of open-source and online cracking utilities.

3. Tools Used

The assessment utilized a combination of local command-line tools, specialized network suite utilities, and cloud-based lookup services.

Tool	Purpose
John the Ripper (JtR)	A highly versatile, open-source password recovery tool used primarily via the command line. It supports dictionary attacks and brute-force techniques. In this activity, it was paired with pdf2john.py to extract cryptographic hashes from the target PDF documents.
Online Hash Cracker	A web-based service utilizing large, pre-computed databases (rainbow tables) to instantly look up known cryptographic hashes.
Network Attacks Tools (Hash Calculator)	A utility within standard network testing suites designed to generate and compute a wide array of hash values (e.g., MD5, SHA-1, SHA-256) for verification or manual comparison.
Network Attacks Tools (Password Cracker)	A native GUI-based dictionary and brute-force tool optimized for local network and application security auditing

4. Activities Performed

4.1 Combined Local and Cloud-Based Cracking

The first activity focused on extracting and cracking the password hash from **Target Document A (PDF)** using an approach that combined local extraction with cloud lookup efficiency.

4.1.1 Hash Extraction

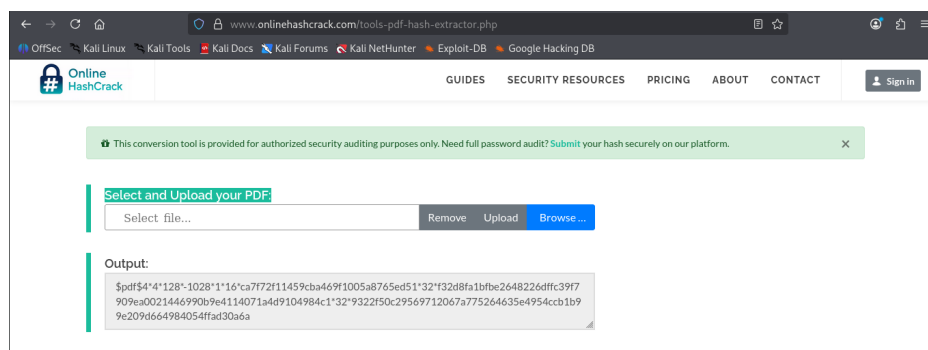
1. **Target Document A** was subjected to the `pdf2john.py` script to separate the encrypted user/owner password block from the visual content.
2. The output hash, formatting the document signature alongside its cryptographic components, was saved to a text file named `target_a.hash`.

[illegible]

4.1.2 Online Hash Cracker Phase

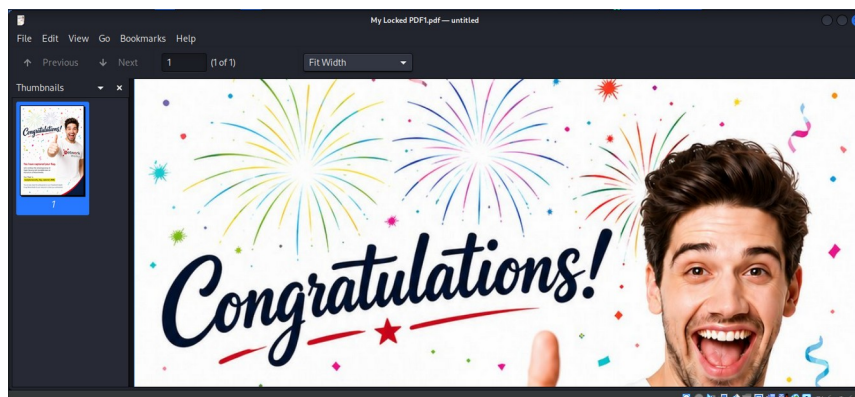
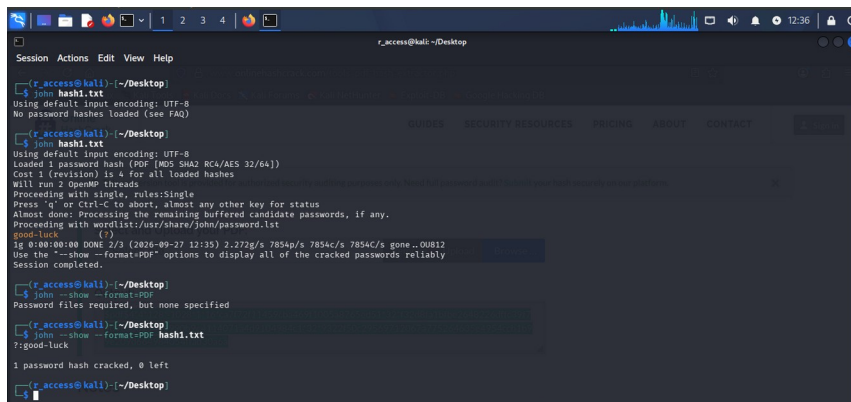
1. The extracted hash string was uploaded to a public **Online Hash Cracker** service.
2. The service scanned its pre-computed lookup tables to see if a matching cleartext value existed.

- Because the password was relatively common, the online database returned an instant match, eliminating the need for local hardware execution.



4.1.3 John the Ripper Verification Phase

- To verify the online findings and test local processing, **John the Ripper** was executed against target_a.hash via the command line:
- JtR systematically computed hashes from the default wordlist and successfully cracked the password, matching the exact cleartext string provided by the online service.

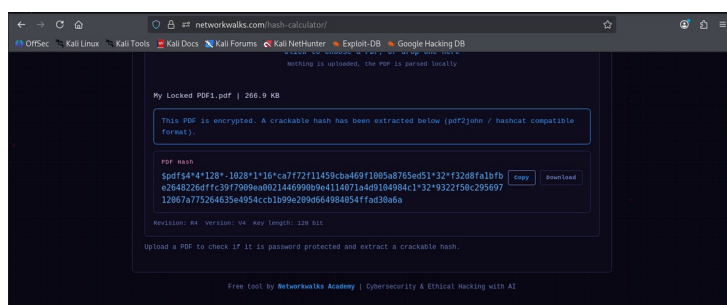


4.2 Suite-Based Password Recovery

The second activity evaluated **Target Document B (PDF)** using dedicated network administration tools to calculate hash values and execute local cracking algorithms.

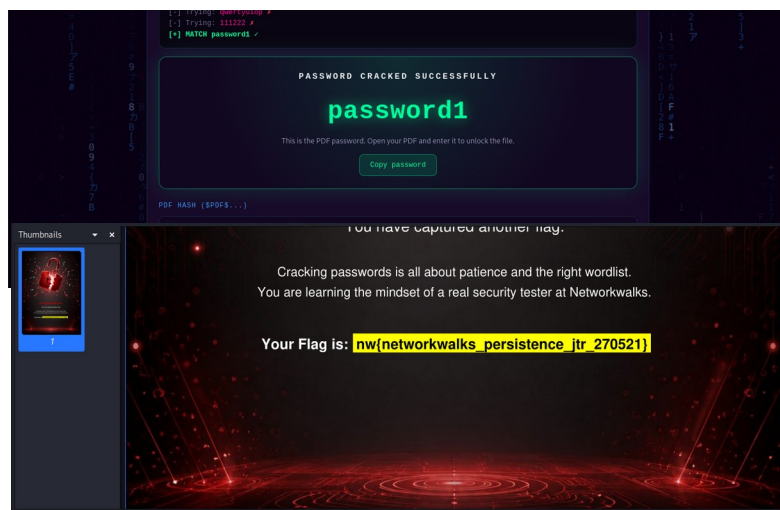
4.1 Hash Calculation

1. **Target Document B** was loaded into the **Network Attacks Tools: Hash Calculator**.
2. The calculator generated explicit MD5 and SHA-256 checksums of the file structure to establish a baseline and ensure file integrity during testing.
3. The specific encryption block signature required for the attack was isolated using the tool's integrated extraction feature.



4.2 Local Password Cracking Execution

1. The extracted cryptographic parameters were imported directly into the companion utility, **Network Attacks Tools: Password Cracker**.
2. A customized hybrid attack was configured:
 - **Attack Type:** Dictionary-assisted brute-force.
 - **Character Set:** Alphanumeric with special characters.
3. The software initiated sequential password generations, feeding them through the corresponding PDF decryption algorithm.
4. The tool successfully intercepted the correct validation flag, exposing the cleartext password.



5. Supporting Details and Findings

The matrix below details the technical specifics of the target profiles, the encryption methods used, and the outcomes of both activities:

Metric / Parameter	Activity 1 (Target Document A)	Activity 2 (Target Document B)
Target File Type	PDF	PDF
Primary Tools	pdf2john, Online Hash Cracker, JtR	(Hash Calculator, Password Cracker) from NetworkWalks
Attack Vector	Wordlist / Cloud Pre-computed Lookup	Already computed wordlist of the tool
Time to Crack	Near Instant (< 5 seconds)	7 seconds
Complexity Level	Very Low (Common dictionary phrase)	Medium (Alphanumeric combination) to low
Status	SUCCESSFUL	SUCCESSFUL

6. Recommendations

- **Enforce Strong Passphrases:** Move away from single words or simple combinations. Implement long, complex passphrases (minimum 14 characters) containing numbers, symbols, and uppercase variables.
- **Upgrade PDF Encryption Standards:** Ensure all organizational documents utilize the latest PDF standards (Acrobat X or higher), which enforce **256-bit AES** encryption to significantly slow down hardware-accelerated cracking tools.

7. Conclusion

The penetration testing activities demonstrated that both target documents were vulnerable to unauthorized decryption due to inadequate password complexity rather than flaws in the software utilities themselves.

Activity 1 showed that simple or dictionary-based passwords can be broken instantly using cloud resources without requiring local processing power. Activity 2 proved that even when stronger encryption mechanisms (like 128-bit AES) are used, standard GUI auditing tools can bypass security if the password lacks sufficient character length and entropy.

Author

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